

ODD-CYCLE COLOURINGS OF ABELIAN CAYLEY GRAPHS COME FROM CHARACTERS

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ABSTRACT. We show that a Cayley graph of an abelian group has a homomorphism to the odd cycle C_{2k+1} if and only if some character maps every generator into the closed arc $[k/(2k+1), (k+1)/(2k+1)]$ of $\mathbb{R}/\mathbb{Z}$; in particular it is 3-colourable if and only if some character maps every generator into $[1/3, 2/3]$. The proof averages the discrete winding number of Krebs and Sankar over the group with an invariant mean. Consequences: (i) in every abelian group, every set of Bohr recurrence is a set of 3-chromatic recurrence, the case of three colours of Katznelson’s question; for every 3-colouring $\mathbb{N} = A_1 \cup A_2 \cup A_3$ the union $\bigcup_i (A_i - A_i)$ contains $\{n : \|n\alpha\| < 1/3\}$ for some α , which answers a question of Glasscock, Koutsogiannis and Richter; (ii) a 3-colourable Cayley graph of a finitely generated abelian group on a finite connection set, for instance a Cayley graph of $\mathbb{Z}^d$, has a periodic 3-colouring, and 3-colourability of these graphs is decidable, while for many colours, by the theorem of Greenfeld and Tao on tilings, some Cayley graphs of $\mathbb{Z}^d$ have no periodic optimal colouring; (iii) Payan’s theorem on cube-like graphs and the exponent-4 case of the theorem of Krebs and Sankar; (iv) a distance graph $G(\mathbb{Z}, D)$ is 3-colourable if and only if $\kappa(D) \geq 1/3$; (v) for a number field F , the plane F^2 is 3-colourable if and only if a character of the group generated by the unit vectors maps them into $[1/3, 2/3]$. For $F = \mathbb{Q}(\sqrt{d})$ and finitely many unit vectors the last condition is a question on a 4-dimensional torus, which exact branch-and-bound certificates settle. We obtain $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 83, 107, 203$, $4 \leq \chi(\mathbb{Q}(\sqrt{143})^2) \leq 5$ and $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$: five fields for which searches for finite non-3-colourable unit-distance graphs had failed.

1. INTRODUCTION

For an abelian group Γ and a subset $S = -S$ of $\Gamma \setminus \{0\}$, the Cayley graph $\text{Cay}(\Gamma, S)$ has vertex set Γ and an edge between g and $g + s$ for $g \in \Gamma$, $s \in S$. A *character* is a homomorphism $\xi : \Gamma \rightarrow \mathbb{R}/\mathbb{Z}$. For $x \in \mathbb{R}/\mathbb{Z}$ let $\|x\|$ be its distance to 0.

Theorem 1. *Let $k \geq 1$ and $n = 2k + 1$. The graph $\text{Cay}(\Gamma, S)$ has a homomorphism to the cycle C_n if and only if there is a character ξ of Γ with $\xi(s) \in [k/n, (k+1)/n]$ for every $s \in S$. In particular, $\text{Cay}(\Gamma, S)$ is 3-colourable if and only if some character maps every element of S into $[1/3, 2/3]$.*

A homomorphism to C_{2k+1} is a circular $(2k+1)/k$ -colouring. Let $\kappa(S) = \sup_\xi \inf_{s \in S} \|\xi(s)\|$; the supremum is attained, as the dual group is compact and $\xi \mapsto \inf_s \|\xi(s)\|$ is upper semicontinuous. The theorem says

$$\chi_c(\text{Cay}(\Gamma, S)) \leq 2 + \frac{1}{k} \iff \kappa(S) \geq \frac{k}{2k+1} \quad (k \geq 1).$$

The *if* direction is the classical inequality $\chi_c \leq 1/\kappa$: for distance graphs it is Zhu’s multiplier method, surveyed by Liu [24, §2], and for Cayley graphs of $\mathbb{Z}/p$ it appears in [3, Theorem 3.2]. The content of the theorem is the converse at the thresholds $2 + 1/k$, among them $\chi \leq 3$.

Steps 1 and 2 of the proof are the discrete winding number of Krebs and Sankar [23] (their Definition 3.5, Propositions 3.3 and 3.6, Remark 3.7, and the parity argument in the proof of

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their Theorem 4.1), which they use to show that groups of exponent 4 have no 3-chromatic Cayley graphs. What is new, as far as we know, is Step 3, which averages the winding over the group, and the characterisation that results (Section 7).

For $k = 1$ the theorem has a reading in terms of recurrence. A set $S \subseteq \Gamma$ is a set of *r-chromatic recurrence* if every r -colouring of Γ has two points of one colour that differ by an element of S , and a set of *Bohr recurrence* if it meets every Bohr neighbourhood of 0, that is, every set $\{g : \|\xi_1(g)\| < \varepsilon, \dots, \|\xi_m(g)\| < \varepsilon\}$ with characters ξ_j and $\varepsilon > 0$ [18, Definition 1.2]. Katznelson's question [21, 15, 18] asks whether every set of Bohr recurrence is a set of r -chromatic recurrence for every r . It is open, and the answer is not known for any countably infinite abelian group [18]. For $r = 2$ the answer is easily yes, and Alweiss notes that a counterexample needs at least three colours [2, §3]. Theorem 1 gives $r = 3$:

Corollary 2. *A set $S \subseteq \Gamma \setminus \{0\}$ is a set of 3-chromatic recurrence if and only if it meets the set $\{g : \|\xi(g)\| < 1/3\}$ for every character ξ . In particular, every set of Bohr recurrence is a set of 3-chromatic recurrence, and every 3-colourable abelian Cayley graph has a 3-colouring whose colour classes are $\xi^{-1}[0, 1/3)$, $\xi^{-1}[1/3, 2/3)$ and $\xi^{-1}[2/3, 1)$ for a single character ξ . For every 3-colouring $\mathbb{N} = A_1 \cup A_2 \cup A_3$ there is $\alpha \in \mathbb{R}/\mathbb{Z}$ with*

$$(A_1 - A_1) \cup (A_2 - A_2) \cup (A_3 - A_3) \supseteq \{n \in \mathbb{N} : \|n\alpha\| < 1/3\};$$

in particular this union is a Bohr₀ set.

Proof. Apply Theorem 1 with $k = 1$ to $S \cup (-S)$; the colouring is the one in the *if* direction of the proof. For the last statement let $S = \mathbb{Z} \setminus \bigcup_i (A_i - A_i)$; then $S = -S$, since each $A_i - A_i$ is symmetric, and $0 \notin S$, since some A_i is nonempty. Every finite subgraph of $\text{Cay}(\mathbb{Z}, S)$ has a translate inside $\mathbb{N}$, where the given colouring is proper, so $\text{Cay}(\mathbb{Z}, S)$ is 3-colourable [9], and Theorem 1 gives α with $\|s\alpha\| \geq 1/3$ for all $s \in S$. $\square$

The radius $1/3$ is best possible. For irrational α , the colouring $n \mapsto \lfloor 3\{n\alpha\} \rfloor$ of $\mathbb{N}$ has $\bigcup_i (A_i - A_i) \cap \mathbb{N} = U := \{n \geq 1 : \|n\alpha\| < 1/3\}$, and U contains no set $\{n \geq 1 : \|n\beta\| < \delta\}$ with $\delta > 1/3$. Indeed, let H be the closure of the multiples of (α, β) in $(\mathbb{R}/\mathbb{Z})^2$; the multiples with $n \geq 1$ are dense in H , so it suffices to find $(x, y) \in H$ with $\|x\| > 1/3$ and $\|y\| < \delta$. If $H = (\mathbb{R}/\mathbb{Z})^2$, take $(1/2, 0)$. Otherwise $H = \{(x, y) : ax + by = 0\}$ for a generator (a, b) of the annihilator of H in $\mathbb{Z}^2$, and $b \neq 0$ because α is irrational. If $|b| \geq 2$, some $(1/2, y) \in H$ has $\|y\| \leq 1/4$. If $b = \pm 1$, then $H = \{(x, cx)\}$ for an integer c , and $x = 1/3 + \varepsilon$ with small $\varepsilon > 0$ has $\|cx\| \leq 1/3 + |c|\varepsilon < \delta$.

For $\Gamma = \mathbb{Z}$ the invariant mean can be avoided. If S is finite with largest element L , two windows $[i, i + L)$ and $[j, j + L)$, $i < j$, of a 3-colouring of $\text{Cay}(\mathbb{Z}, S)$ carry the same colours, and repeating the colouring on $[i, j)$ with period $N = j - i$ gives a 3-colouring of $\text{Cay}(\mathbb{Z}/N, S)$; Theorem 1 for the finite group $\mathbb{Z}/N$, where the mean is a plain average, gives $\alpha \in \frac{1}{N}\mathbb{Z}/\mathbb{Z}$. For infinite S , a limit point of the characters obtained for finite $S_m \uparrow S$ works, since the conditions $\|s\alpha\| \geq 1/3$ are closed. So the last statement of Corollary 2 needs only the finite cyclic case of Theorem 1.

The last statement answers Question 3 of Glasscock, Koutsogiannis and Richter [15], who proved it for two colours [15, Theorem 4.9] and wrote that the case of three colours was not known. As they remark, it follows that the difference set of a set $A \subseteq \mathbb{N}$ with $A \cup (A - \ell_1) \cup (A - \ell_2) \supseteq \mathbb{N}$ is a Bohr₀ set. In the same way, if three translates of $A \subseteq \mathbb{Z}^d$ cover $\mathbb{Z}^d$, then $A - A$ contains $\{z : \|z \cdot \lambda\| < 1/3\}$ for some $\lambda \in (\mathbb{R}/\mathbb{Z})^d$, the case of three translates of [15, Question 8]. By Corollary 2, a counterexample to Katznelson's question needs at least four colours. We have not found the case $r = 3$ in [15, 18, 19, 25, 2], and a separate agent's independent search found nothing either; we have not been able to read [21].

A second consequence concerns periodicity. A colouring of a Cayley graph is *periodic* if it is invariant under a subgroup of finite index.

Corollary 3. *Let Γ be a finitely generated abelian group and S finite. If $\text{Cay}(\Gamma, S)$ has a homomorphism to C_{2k+1} (for $k = 1$: if it is 3-colourable), then it has a periodic one, and whether it has one is decidable.*

Proof. Write $\Gamma \cong \mathbb{Z}^r \times F$ with F finite. In these coordinates the characters with $\xi(S) \subseteq [k/n, (k+1)/n]$ form a finite union of polytopes with rational vertices in the torus $(\mathbb{R}/\mathbb{Z})^r \times \widehat{F}$. If one exists, a rational one exists; it has finite order, and the colouring of the *if* direction of the proof is constant on the cosets of its kernel, a subgroup of finite index. The vertices can be enumerated. $\square$

For $\Gamma = \mathbb{Z}$ and finite S (distance graphs) there are periodic optimal colourings for any number of colours, by a pigeonhole argument. For $\mathbb{Z}^d$ with $d \geq 2$ this argument fails, and subshifts of finite type on $\mathbb{Z}^2$ can be aperiodic. Abrishami, Esperet, Giocanti, Hamann, Knappe and Möller ask whether every locally finite graph with a periodic proper colouring has a periodic proper colouring with $\chi(G)$ colours [1, Problem 4.6]; they answer yes for graphs of bounded pathwidth, such as Cayley graphs of 2-ended groups, and no for some ∞ -ended graphs. Corollary 3 answers yes for Cayley graphs of finitely generated abelian groups with finite connection sets and $\chi \leq 3$ (if $\chi = 2$, the parity of $\text{Cay}(\Gamma', S)$ is a character of order 2 of the subgroup Γ' generated by S ; extended to a character of finite order of Γ , it gives a periodic 2-colouring). Vallentin, Weißbach and Zimmermann note that it is not known whether the chromatic number of a lattice, that is, of the Cayley graph of $\Lambda \cong \mathbb{Z}^n$ on its strict Voronoi vectors, is computable, and ask whether a periodic colouring with $\chi(\Lambda)$ colours always exists [31]. By Corollary 3, whether $\chi(\Lambda) \leq 3$ is decidable (given the strict Voronoi vectors in coordinates of Λ), and if it is, a periodic 3-colouring exists. Three is special: by the theorem of Greenfeld and Tao on the periodic tiling conjecture [16], some Cayley graphs of $\mathbb{Z}^d$ have no periodic colouring with χ colours (Proposition 11).

Section 2 proves the theorem. Section 3 derives Payan's theorem [28] and the exponent-4 case of Krebs and Sankar [23]. Section 4 treats distance graphs. Section 5 applies the theorem to the unit-distance graph of the plane over a number field, the setting that led to it, and Section 6 records that nothing of the kind holds for four colours, and that periodicity fails for many colours.

2. PROOF OF THEOREM 1

If. Let $c(g) = 2\lfloor n\xi(g) \rfloor \bmod n$, where $\xi(g)$ is read in $[0, 1)$. Along an edge, $n\xi$ moves by an amount in $[k, k+1]$ modulo n , so $\lfloor n\xi \rfloor$ moves by k or $k+1$ modulo n , and c moves by $2k \equiv -1$ or $2k+2 \equiv 1$. So c is a homomorphism to C_n , whose vertices are $\mathbb{Z}/n$ with $j \sim j \pm 1$.

Only if. Let $c : \Gamma \rightarrow \mathbb{Z}/n$ be a homomorphism to C_n , and let Γ' be the subgroup generated by S . A character of Γ' extends to Γ , since $\mathbb{R}/\mathbb{Z}$ is divisible, so it suffices to find the character on Γ' . We work in Γ' because $\text{Cay}(\Gamma', S)$ is connected, which Step 2 uses: averaging over all of Γ would fail when $\text{Cay}(\Gamma, S)$ is disconnected, since different components can wind in opposite directions. For $g \in \Gamma'$ and $s \in S$ let $\sigma(g, s) \in \{1, -1\}$ be the lift of $c(g+s) - c(g)$; then $\sigma(g+s, -s) = -\sigma(g, s)$.

Step 1: squares do not wind [23, Proposition 3.6]. For $s, t \in S$ the two integers $\sigma(g, s) + \sigma(g+s, t)$ and $\sigma(g, t) + \sigma(g+t, s)$ lie in $\{-2, 0, 2\}$ and are congruent to $c(g+s+t) - c(g)$ modulo n . Two distinct elements of $\{-2, 0, 2\}$ differ by 2 or 4, which n (odd, at least 3) does not divide; so they are equal.

Step 2: the winding of a closed walk [23, Definition 3.5, Propositions 3.3 and 3.6, Remark 3.7]. Let Z be the abelian group generated by symbols $[s]$, $s \in S$, subject to $[-s] = -[s]$. It is free on one element of each pair $\{s, -s\}$, except that $2[s] = 0$ when $2s = 0$; so the parity of the number of symbols is a well-defined homomorphism $Z \rightarrow \mathbb{Z}/2$. Let R be the kernel of $Z \rightarrow \Gamma'$, $[s] \mapsto s$.

A closed walk from g with steps $s_1, \dots, s_N$ ($\sum s_j = 0$) has the sum $\sum_j \sigma(g+s_1+\dots+s_{j-1}, s_j) \equiv 0 \pmod{n}$. By Step 1 this sum does not change when two consecutive steps are exchanged, nor when a step s immediately followed by $-s$ is deleted. Every walk can be brought to a normal form by these moves (sort the steps by exchanges, then cancel the pairs $s, -s$), so the sum depends only on the base point and on the class $r = \sum_j [s_j] \in R$. It does not depend on the base point either: going from $g+s$ to g , doing the walk and coming back gives the same sum, and $\text{Cay}(\Gamma', S)$ is connected. Write the sum as $nw_c(r)$. Then $w_c : R \rightarrow \mathbb{Z}$ is a homomorphism (concatenate walks). Since $nw_c(r)$ is a sum of N odd numbers and n is odd, $w_c(r)$ has the parity of r .

Step 3: averaging. As Γ' is abelian, it is amenable: there is a mean m on the bounded functions on Γ' that is invariant under translations. For finite Γ' it is the plain average, and for finitely generated Γ' a limit along Følner sets. Let

$$f(s) = m(g \mapsto \sigma(g, s)) \in [-1, 1].$$

Then $f(-s) = -f(s)$. For a closed walk $(s_1, \dots, s_N)$ that represents $r \in R$, the function $g \mapsto \sum_j \sigma(g + s_1 + \dots + s_{j-1}, s_j)$ is the constant $n w_c(r)$ by Step 2. Applying m , which is linear and invariant under translations, gives

$$\sum_{j=1}^N f(s_j) = n w_c(r).$$

Step 4: the character. Put $\xi(s) = \frac{1}{2}(f(s)/n + 1) \in [k/n, (k+1)/n]$; then $\xi(-s) = 1 - \xi(s)$. If $2s = 0$, then $\sigma(g + s, s) = -\sigma(g, s)$ for all g , so $f(s) = 0$ and $\xi(s) = 1/2$. Hence ξ is well defined modulo 1 on Z . For $r \in R$ represented by a walk of length N , $\sum_j \xi(s_j) = \frac{1}{2}(w_c(r) + N)$, an integer by the parity in Step 2. So ξ vanishes on R modulo 1 and defines a character of $Z/R = \Gamma'$, with $\xi(S) \subseteq [k/n, (k+1)/n]$. $\square$

Remark 4. For $\Gamma = \mathbb{Z}^2$ with the standard generators this is the height-function picture of 3-colourings of the square lattice, in which the height changes by ± 1 along edges and by 0 around squares (compare [30]); f is the averaged slope. Theorem 1 says that the averaged slopes of any colouring of any abelian Cayley graph form a character as above. The colouring c itself need not come from a character: for $\Gamma = \mathbb{Z}^2$ most 3-colourings do not.

Remark 5 (Formal proof). For finite Γ and $k = 1$ the *only if* direction is formalised in Lean 4 with Mathlib, in the file `lean/TheoremW.lean` of [11]. The formal proof follows Steps 1–4, with the plain average as the mean; it depends only on Lean’s three standard axioms, and the repository’s continuous integration replays it in Lean’s kernel.

3. GROUPS OF SMALL EXPONENT

Corollary 6. *Let Γ be an abelian group of exponent 2 or 4. Then no Cayley graph of Γ has chromatic number 3.*

Proof. A character of Γ takes values in $\frac{1}{4}\mathbb{Z}/\mathbb{Z}$, and the only such value in $[1/3, 2/3]$ is $1/2$. So if $\text{Cay}(\Gamma, S)$ is 3-colourable, a character maps S to $1/2$, and $g \mapsto \lfloor 2\xi(g) \rfloor$ is a proper 2-colouring. $\square$

For exponent 2 this is Payan’s theorem on cube-like graphs [28], which Cervantes and Krebs reproved [7]. For exponent 4 it is a case of the theorem of Krebs and Sankar [23, Theorem 1.2]: an abelian group has a 3-chromatic Cayley graph if and only if its exponent is not 1, 2 or 4. Their proof uses the same winding number: for exponent 4 they show that the discrete fundamental group is torsion, and by their Theorem 4.1 a non-bipartite graph with a torsion fundamental group is not 3-colourable. The same argument as for Corollary 6 gives, for instance, that a Cayley graph of an abelian group of exponent 8 that maps to C_5 is bipartite, since $1/2$ is the only element of $\frac{1}{8}\mathbb{Z}/\mathbb{Z}$ in $[2/5, 3/5]$. For C_3 the values $3/8$ and $5/8$ are also available, and indeed such groups have 3-chromatic Cayley graphs [23].

4. DISTANCE GRAPHS

For a finite set D of positive integers, the distance graph $G(\mathbb{Z}, D)$ is $\text{Cay}(\mathbb{Z}, \pm D)$. Its characters are $x \mapsto \alpha x$, $\alpha \in \mathbb{R}/\mathbb{Z}$, and $\kappa(D) = \max_{\alpha} \min_{d \in D} \|\alpha d\|$ is the parameter of the lonely runner problem [24, 29].

Corollary 7. *$G(\mathbb{Z}, D)$ is 3-colourable if and only if $\kappa(D) \geq 1/3$; it maps to C_{2k+1} if and only if $\kappa(D) \geq k/(2k+1)$.*

The set of α with $\|\alpha d\| \geq 1/3$ for all $d \in D$ is a finite union of closed intervals whose endpoints are among the points $(j+1/3)/d$ and $(j+2/3)/d$. So it is non-empty if and only if it contains one of these points, and the test is finite. We compared it, for all 1747 sets $D \subseteq [1, 24]$ with $|D| = 3$

and $\gcd(D) = 1$, with 3-colourability of the segment $[0, 12 \max D + 59]$ decided by a SAT solver. They agree, and the 74 sets that need four colours are exactly those of Zhu's classification [32] of 3-element sets with $\chi(G(\mathbb{Z}, D)) = 4$, namely $\{1, 2, 3m\}$ and $\{a, b, a + b\}$ with $a \not\equiv b \pmod{3}$ (after division by the gcd). Liu's survey asks whether $\chi_c(G(\mathbb{Z}, D)) < 1/\kappa(D)$ can happen when $|D| = 3$ [24, Problem 3]. By Corollary 7, if one of the two numbers is at most 3, then both equal 2 or both lie in the same interval $(2 + 1/(k + 1), 2 + 1/k]$.

5. PLANES OVER NUMBER FIELDS

Let F be a field of characteristic 0 with $i \notin F$, and $L = F(i)$. Identify F^2 with L by $(x, y) \mapsto x + iy$; the unit vectors are $T = \{z \in L : z\bar{z} = 1\}$, and the unit-distance graph of F^2 is $\text{Cay}(L, T)$, a disjoint union of copies of $\text{Cay}(A, T)$ with $A = \mathbb{Z}[T]$.

Corollary 8. $\chi(F^2) \leq 3$ if and only if some character of A maps every unit vector into $[1/3, 2/3]$. If for a finite set $U \subset T$ no character of the group $\mathbb{Z}U$ maps U into $[1/3, 2/3]$, then $\chi(F^2) \geq 4$, and some finite unit-distance graph over F is not 3-colourable.

The last assertion is the theorem of de Bruijn and Erdős [9]. If F is a number field, then $\mathbb{Z}U$ is a finitely generated subgroup of the $\mathbb{Q}$ -vector space L , hence free of rank at most $[L : \mathbb{Q}]$; for a quadratic field F the test for a finite U is a question on a torus of dimension at most 4. A locally constant 3-colouring at a place of F [10, 27, 12] is a colouring of a finite quotient of A , and Theorem 1 applied to that quotient gives such a character.

5.1. Real quadratic fields. For $F = \mathbb{Q}(\sqrt{d})$ with $d > 1$ squarefree, $\chi(F^2) = 2$ if $d \not\equiv 3 \pmod{4}$ [20, 10, 27], $\chi(F^2) \leq 3$ if moreover $d \not\equiv 2 \pmod{3}$ [10, 27], and $\chi(F^2) \leq 4$ if $d \equiv 3 \pmod{8}$ [10, Theorem 10] or $d \equiv 0, 1, 2, 4 \pmod{7}$ [27]. In [13] we found finite non-3-colourable unit-distance graphs over $\mathbb{Q}(\sqrt{d})$ for 27 values $d \equiv 11 \pmod{12}$ below 1000, but searches for such graphs failed for $d = 83, 107, 143, 167, 203$, the values that the local-global question of [12] predicted to need four colours and that remained open below 210.

Fix a denominator D and let U_D be the unit vectors $((a + b\sqrt{d})/D, (c + e\sqrt{d})/D)$, $a, b, c, e \in \mathbb{Z}$, one of each pair $\pm u$. A character with $\xi(U_D) \subseteq [1/3, 2/3]$ gives $f \in [1/3, 2/3]^{U_D}$ (f_u the lift of $\xi(u)$ in $[1/3, 2/3]$) with $\langle r, f \rangle \in \mathbb{Z}$ for every integer relation r among the vectors of U_D . Using an LLL-reduced basis of the relations, we rule this out by branch and bound over the integers $\langle r, f \rangle$: each leaf carries a rational vector y (a Farkas certificate) such that $\langle y, \cdot \rangle$ applied to the equations fixed at that leaf cannot hold on the box $[1/3, 2/3]^{U_D}$. A short program checks a certificate in exact integer and rational arithmetic: the vectors have length 1 ($a^2 + db^2 + c^2 + de^2 = D^2$, $ab + ce = 0$), the relations hold, every branch covers every integer value in its range, and every Farkas vector is valid. Any set of valid relations gives necessary conditions, so infeasibility is a proof whether or not the relations form a basis.

Theorem 9. $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 83, 107, 203$; $4 \leq \chi(\mathbb{Q}(\sqrt{143})^2) \leq 5$; and $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$.

Proof. The lower bounds are Corollary 8 with the certificates of Table 1 [11]. The upper bound 4 holds because $83 \equiv 107 \equiv 203 \equiv 3 \pmod{8}$ [10, Theorem 10]; see also [4, Theorem 2.2]. For $d = 143$ the place above 11 is ramified in $\mathbb{Q}(\sqrt{143})$ with residue field $\mathbb{F}_{11}$, and i is not in its completion; unit vectors are integral there and reduce to the circle $x^2 + y^2 = 1$ of $\mathbb{F}_{11}^2$, whose graph is 5-colourable [13], which colours each coset of the valuation ring. $\square$

With the values of [13], $\chi(\mathbb{Q}(\sqrt{d})^2)$ is now known for every squarefree $d < 143$ except $d = 47$. $\mathbb{Q}(\sqrt{167})$ is the first admissible field of [12]: no place has a locally constant 4-colouring, so $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$ is also the three-colour case of the local-global question there. For $d = 203$ the result shows that no level of the place above 7 is 3-colourable, which [12] had checked only up to level 3. For $d = 83$ and $D = 510$, colouring-guided growth had stopped at 6008 points because a 3-colouring of the ball of radius 2 extended to all 215478 candidate points, so the obstruction does not lie in a small ball. A subset of 28 of the 54 vectors, found by greedy deletion, already suffices for $d = 83$. For $\mathbb{Q}(\sqrt{11})$, a subset of 16 vectors with $D = 30$ suffices, in agreement with the 76-vertex graph of [13].

d	D	vectors	relations	nodes	leaves
83	510	54	50	6 886	4 664
107	1170	90	86	12 785	9 618
143	1740	54	50	3 861	2 710
167	1560	54	50	8 552	6 094
203	1530	78	74	1 072	732

TABLE 1. Certificates that no character of $\mathbb{Z}U_D$ maps U_D into $[1/3, 2/3]$ (vectors counted up to sign). In each case the vectors span a group of rank 4 and the relations listed generate all relations.

The lower bounds rest on Theorem 1, whose averaging step is not constructive, and on the certificates; we give no explicit finite graph for the five fields. We expect such graphs to be large, which would explain why the searches failed, but this is speculation: Step 3 says nothing about their size. A quantitative version of Step 3 over Følner boxes, or an explicit certified graph, would remove the dependence on the infinite averaging.

All five denominators are even, as they must be. For odd D every vector of U_D has $a + b + c + e$ odd, because $a^2 + db^2 + c^2 + de^2 = D^2$ and d is odd; so $x \mapsto \frac{1}{2}(a + b + c + e)$, for $Dx = (a + b\sqrt{d}) + i(c + e\sqrt{d})$, is a character of $\mathbb{Z}U_D$ with the value $1/2$ on U_D , and $\text{Cay}(\mathbb{Z}U_D, U_D)$ is bipartite. The infeasible denominators we found are all multiples of 6, but some multiples of 6 are feasible (for instance $D = 1170$ for $d = 203$). As controls, the test is feasible for 40 pairs (d, D) with $d \in \{2, 3, 5, 6, 7, 15, 19, 31, 39, 43\}$, whose planes are known to be 3-colourable, and for the even denominators $(d, D) = (107, 1450)$, $(143, 1752)$, $(167, 1820)$ and $(203, 1590)$, whose unit vectors also span a group of rank 4.

Question 10. Is $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every squarefree $d \equiv 11 \pmod{12}$? In terms of Corollary 8 and [12]: for real quadratic fields, does a character that maps all unit vectors into $[1/3, 2/3]$ exist only when a place gives a locally constant 3-colouring?

6. FOUR AND MORE COLOURS

Step 1 fails for K_4 , and so does the theorem: $K_4 = \text{Cay}((\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \setminus \{0\})$ is 4-colourable, but every character of $(\mathbb{Z}/2)^2$ vanishes on some nonzero element, so none maps the three generators into $[1/4, 3/4]$. The inequality $\chi_c \leq 1/\kappa$ holds for all k , so a character with $\xi(S) \subseteq [1/4, 3/4]$ still gives a 4-colouring. For the 108 unit vectors of $\mathbb{Q}(\sqrt{47})$ with denominator 240, which underlie the search for a five-chromatic triangle-free unit-distance graph in [12], a floating-point computation finds none; we have not certified this.

For many colours even Corollary 3 fails, because colourings can encode tilings.

Proposition 11. *There are a finitely generated abelian group Γ (one can take $\Gamma = \mathbb{Z}^d$ for some d , or $\Gamma = \mathbb{Z}^2 \times G_0$ with G_0 finite) and a finite symmetric $S \subseteq \Gamma \setminus \{0\}$ such that $\text{Cay}(\Gamma, S)$ has a proper colouring with $k = \chi(\text{Cay}(\Gamma, S)) < \infty$ colours but no periodic one.*

Proof. For a finite $F \subseteq \Gamma$ let $S_F = (F - F) \setminus \{0\}$. Every translate $x + F$ is a clique of $\text{Cay}(\Gamma, S_F)$ with $|F|$ vertices, so $\chi \geq |F|$, and in a proper colouring with $|F|$ colours every colour class A meets every translate $x + F$ exactly once, that is, $A \oplus (-F) = \Gamma$. Conversely, if $F \oplus A = \Gamma$, giving $f + a$ the colour f is a proper colouring. A periodic colouring gives a periodic tiling (a union of finitely many cosets of a subgroup of finite index), and conversely. Greenfeld and Tao [16, Theorem 1.4 and Corollary 1.6] construct a finite F that tiles $\mathbb{Z}^2 \times G_0$ for some finite abelian G_0 , and one that tiles $\mathbb{Z}^d$ for some d , such that no tiling by F is periodic. Take $S = S_F$ and $k = |F|$. $\square$

For $\Gamma = \mathbb{Z}^d$ one can also make S generate: replace $\mathbb{Z}^d$ by the subgroup generated by $F - F$, which is free abelian, and note that a periodic tiling of it would extend to a periodic tiling of $\mathbb{Z}^d$. Every automorphism of a connected Cayley graph of $\mathbb{Z}^d$ with a finite connection set is affine [26], so a colouring whose colour-preserving automorphisms act with finitely many orbits is

invariant under a subgroup of finite index; and $x \mapsto x \bmod N\mathbb{Z}^d$ is a periodic proper colouring once $S \cap N\mathbb{Z}^d = \emptyset$. So the answer to [1, Problem 4.6] is no for Cayley graphs of $\mathbb{Z}^d$, while Corollary 3 gives yes when $\chi \leq 3$. We do not know the least k in Proposition 11; it is at least 4. Nor do we know whether every k -colourable Cayley graph of $\mathbb{Z}^2$ has a periodic k -colouring: a single tile of $\mathbb{Z}^2$ always tiles periodically [5], so the examples above need torsion or $d \geq 3$. Nor do we know whether k -colourability of Cayley graphs of finitely generated abelian groups is decidable for $k \geq 4$: Greenfeld and Tao [17] proved that tiling a periodic subset of $\mathbb{Z}^2 \times G_0$ by one tile is undecidable, but a colouring problem cannot single out a subset. The question of [31] concerns the Voronoi connection set and is not touched.

7. LITERATURE

The closest work is that of Krebs and Sankar [23], whose winding number is Steps 1 and 2; they use it for torsion fundamental groups, without characters or averaging. The *if* direction of Theorem 1 is classical [24, 3, 14]. The following are consistent with Theorem 1 and do not imply it: Cervantes and Krebs [6, 7, 8] (in [8], 3-colourability for small dimension and rank is characterised by two forbidden subgraphs, diamond lanyards and $C_{13}(1, 5)$, and $C_{13}(1, 5)$ indeed has no suitable character); Krebs and Leyva [22]; the work on Katznelson’s question on Bohr recurrence [18, 19, 25], where we found no statement for three colours (we have not seen [21] itself); and Perarnau and Serra [29, §3.5], which has $\chi_f \leq \chi_c \leq 1/\kappa$. We did not find Theorem 1 in these works, in the literature on distance graphs and the lonely runner problem through Liu’s survey [24], or in web searches on characters, circular colourings and odd-cycle homomorphisms of Cayley graphs. The proof uses only classical tools, so an earlier occurrence is possible.

By Theorem 1, whether a finite abelian Cayley graph, given by Γ and S , maps to C_{2k+1} can be decided by checking the $|\Gamma|$ characters, with $O(|\Gamma| |S|)$ operations: polynomial in the size of the graph.

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